with $\mathbf{T}^0 = -\mathbf{Te}$ being the transition intensity vector from a transient state up to one absorbing state.

It can be seen that if $\boldsymbol{\alpha}$ is the scalar 1 and $\mathbf{T}$ is the scalar $-\lambda$, the exponential distribution is achieved. The reliability function, R(v), describes the probability that at voltage v the conductive filament is not broken, and it is given by Equation 3

$$R(v) = 1 - F(v) = \boldsymbol{\alpha} \exp(\mathbf{T}v)\mathbf{e}\,,\; v \geq 0. \tag{3}$$

Thus, the cumulative hazard rate is given by Equation 4

$$H(v) = -\ln(1 - F(v)) = -\ln(\boldsymbol{\alpha} \exp(\mathbf{T}v)\mathbf{e}), \tag{4}$$

and then the hazard rate is

$$h(v) = \frac{\partial H(v)}{\partial v} = \frac{f(v)}{1 - F(v)} = \frac{\boldsymbol{\alpha} \exp(\mathbf{T}v)\mathbf{T}^0}{\boldsymbol{\alpha} \exp(\mathbf{T}v)\mathbf{e}}\,,\; v \geq 0.$$

3.2. Some PHD Properties

Phase-type distributions are important not only because of their structure but also for the good properties which enable to ease the applicability and interpretation of results.

Many well known distributions, in addition to the exponential distribution mentioned above, are PHD. Next, some of these are exposed with the corresponding PHD representation.

1. The Erlang distribution $F(v) = 1 - \sum_{j=0}^{m-1} e^{-\lambda v} (\lambda v)^j / j!$ for $v \geq 0$, $m \geq 1$ and $\lambda > 0$,

$$\boldsymbol{\alpha} = (1, \ldots, 0, 0),\; \mathbf{T} = \begin{pmatrix} -\lambda & \lambda & & \\ & -\lambda & \ddots & \\ & & \ddots & \lambda \\ & & & -\lambda \end{pmatrix}_{mxm}.$$

2. Hypo-exponential distribution $F(v) = 1 - \sum_{x=0}^{v} \sum_{i=1}^{m} \lambda_i e^{-\lambda_i x} \left(\prod_{\substack{j=1 \\ j \neq i}}^{m} \frac{\lambda_j}{\lambda_j - \lambda_i} \right)$ for $v \geq 0$, $\lambda_i \neq \lambda_j$ for $i \neq j$,

$$\boldsymbol{\alpha} = (1, 0, \ldots, 0), \ \mathbf{T} = \begin{pmatrix} -\lambda_1 & \lambda_1 & & \\ & -\lambda_2 & \ddots & \\ & & \ddots & \lambda_{m-1} \\ & & & -\lambda_m \end{pmatrix}.$$

3. Hyper-exponential distribution $F(v) = 1 - \sum_{i=1}^{m} \alpha_i \left(1 - e^{-\lambda_i v}\right)$ for $v \geq 0$,

$$\boldsymbol{\alpha} = (\alpha_1, \alpha_2, \ldots, \alpha_m), \ \mathbf{T} = \begin{pmatrix} -\lambda_1 & & & \\ & -\lambda_2 & & \\ & & \ddots & \\ & & & -\lambda_m \end{pmatrix}$$

4. Coxian distribution

$$\boldsymbol{\alpha} = (1, 0, \ldots, 0), \ \mathbf{T} = \begin{pmatrix} -\lambda_1 & g_1\lambda_1 & & \\ & -\lambda_2 & g_2\lambda_2 & \\ & & \ddots & g_{m-1}\lambda_{m-1} \\ & & & -\lambda_m \end{pmatrix}$$

5. Generalized Coxian distribution

$$\boldsymbol{\alpha} = (\alpha_1, \alpha_2, \ldots, \alpha_m), \ \mathbf{T} = \begin{pmatrix} -\lambda_1 & g_1\lambda_1 & & \\ & -\lambda_2 & g_2\lambda_2 & \\ & & \ddots & g_{m-1}\lambda_{m-1} \\ & & & -\lambda_m \end{pmatrix}$$

In general, the following result [6] describes when a non-negative probability distribution is PHD. Then, a non-negative probability distribution is a phase-type distribution if and only if it is either the point mass at zero or; it has a strictly positive continuous density on the positive real numbers and it has a rational Laplace-Stieltjes transform with a unique pole of maximal real part.

One essential property that verifies the phase-type distribution class is that this class is dense in the set of probability distributions on the non-negative half-line [25]. In this way, PHDs can be considered general distributions with a well-structured matrix algorithmic form. On the other hand, any non-negative probability distribution can be approximated as much as desired by a phase-type-distribution.

Other properties for phase-type distributions are the closure properties. The PHD class is closed under a number of operations such as minimum, maximum and addition.

3.3. Estimating phase-type distributions: the EM algorithm

Fitting PHDs is a difficult optimization problem given that the representation of a PHD is highly redundant in general. One usual technique used to estimate the parameters of a PHD is the Expectation Maximization (EM) algorithm. The first EM algorithm was developed by [38] and assumed by [39].

Let $v_1,\ldots,v_n$ be a sequence of n observed variable values. In our case, the value v_i is the voltage up to the absorption (rupture of the filament). The set $\{v_1,\ldots,v_n\}$ includes the outcomes of n independent replications of a PHD with representation $(\boldsymbol{\alpha},\mathbf{T})$ associated to an absorbing Markov process. The likelihood function to be optimized in the EM algorithm can be expressed as

$$L(\boldsymbol{\alpha},\mathbf{T})=\prod_{i=1}^{m}\boldsymbol{\alpha}_i^{N_i}\prod_{i=1}^{m}e^{x_i\mathbf{T}_{ii}}\prod_{i=1}^{m}\prod_{\substack{j=1\\ j\neq i}}^{m+1}\mathbf{T}_{ij}^{N_{ij}} \ ,$$

where N_i is the number of times that the Markov process started in phase i, x_i is the total time spent in phase i and N_{ij} is the total observed number of jumps between both states, i and j.